DIGITAL IMAGES

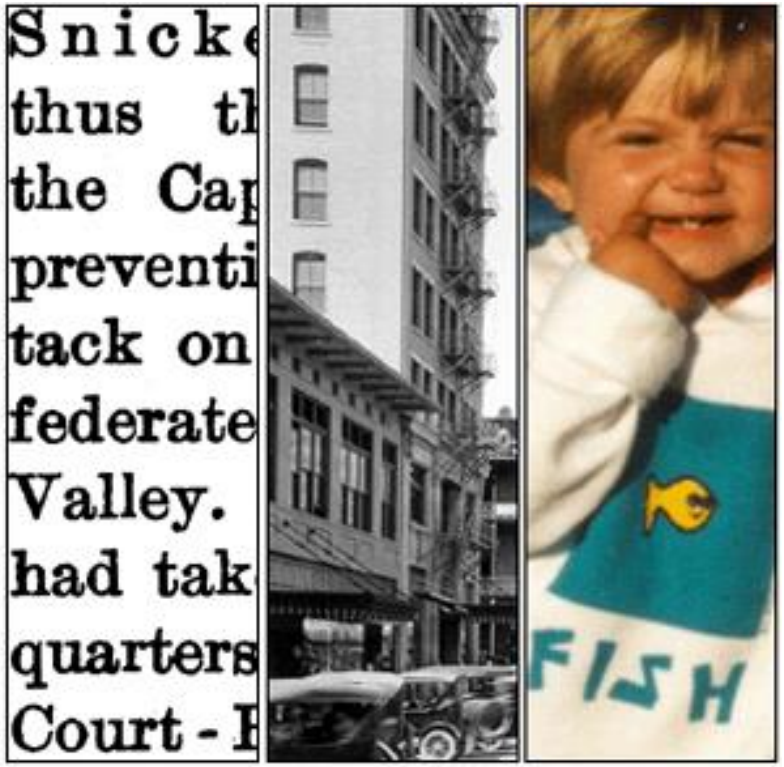
Digital images

- Digital images are electronic snapshots taken of a scene or scanned from documents, such as photographs, manuscripts, printed texts, and artwork

- The digital image is sampled and mapped as a grid of dots or picture elements (pixels). Each pixel is assigned a tonal value (black, white, shades of gray or color), which is represented in binary code
- The bits for each pixel are stored in a sequence by a computer and often reduced to a mathematical representation (compressed)

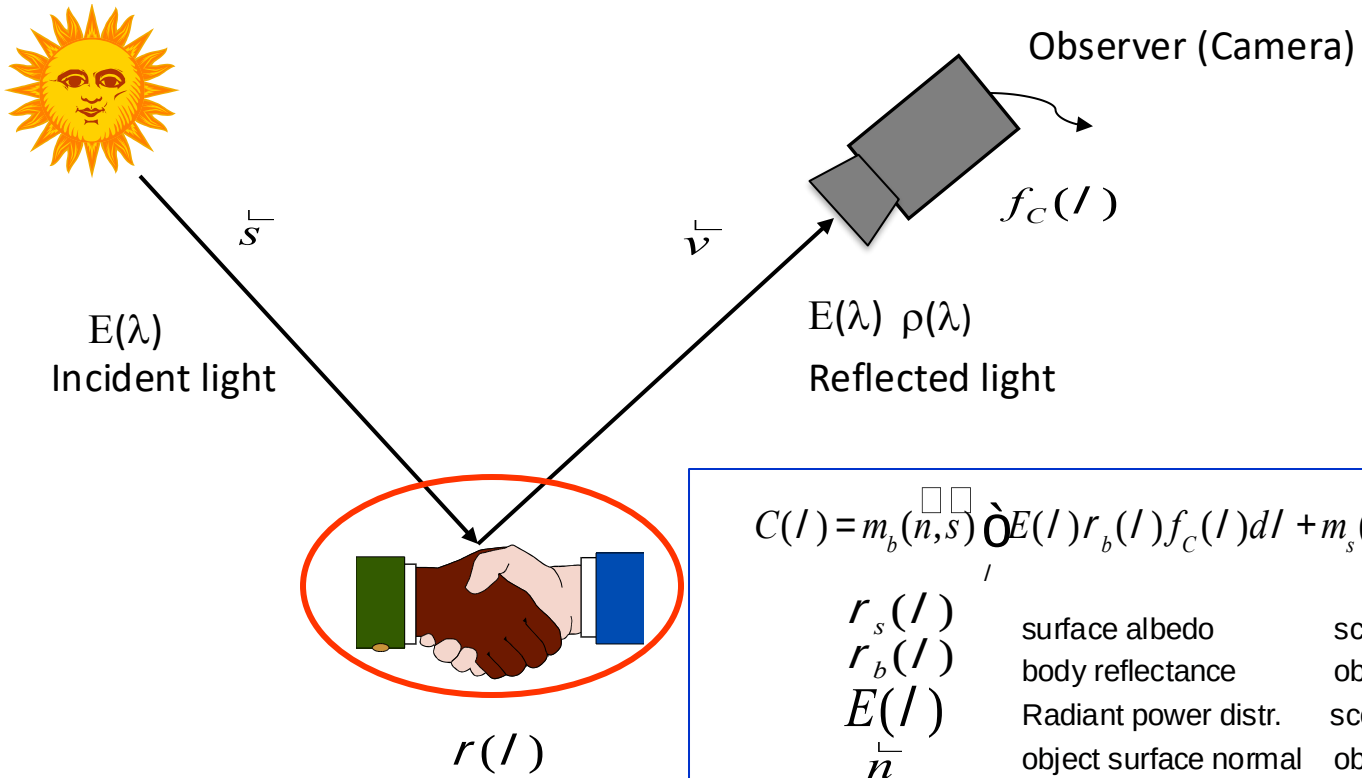
Bit depth

- Bit depth is determined by the number of bits used to define each pixel
  - A *bitonal image* is represented by pixels consisting of 1 bit each, which can represent two tones (typically black and white)
  - A *grayscale image* is composed of pixels represented by multiple bits of information, typically ranging from 2 to 8 bits or more
  - A *color image* is typically represented by a bit depth ranging from 8 to 24 or higher



Color image formation

- Color image formation is determined by the relative radiant power distribution of the incident light, the reflection of the materials and the characteristics of the camera observer.



$$C(\lambda) = m_b(\vec{n},\vec{s}) \int_{\lambda} E(\lambda) r_b(\lambda) f_C(\lambda) d\lambda + m_s(\vec{n},\vec{s},\vec{v}) \int_{\lambda} E(\lambda) r_s(\lambda) f_C(\lambda) d\lambda$$

$r_s(\lambda)$	surface albedo	scene & viewpoint invariant
$r_b(\lambda)$	body reflectance	object material dependent
$E(\lambda)$	Radiant power distr.	scene dependent
$\vec{n}$	object surface normal	object shape variant
$\vec{s}$	illumination direction	scene dependent
$\vec{v}$	viewer's direction	viewpoint variant
$f_C(\lambda)$	sensor sensitivity	device dependent

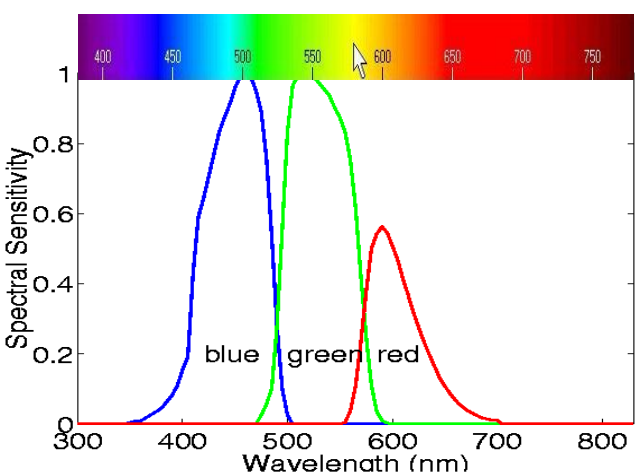
Digital RGB cameras

- Digital images are nowadays obtained from photographic digital cameras that use CMOS or CCD sensors to acquire three color signals in the *red* (R) *green* (G) and *blue* (B) wavelenghts.

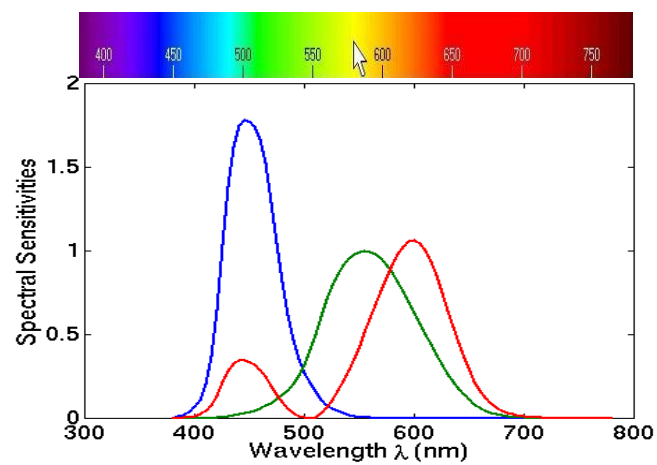
- Motivations for RGB sensors originate from mimicking the way in which human eyes (the cone cells of the retina) operate. The reflected light spectrum can be represented by a 3-element vector, with values which are the proportions of each of the primary colors *red* (R), *green* (G) and *blue* (B) used to produce it. These are the *tristimulus* values. Considering Tristimulus:

RGB values of camera = Colour * Tristimulus.

Camera Response



Eye Response

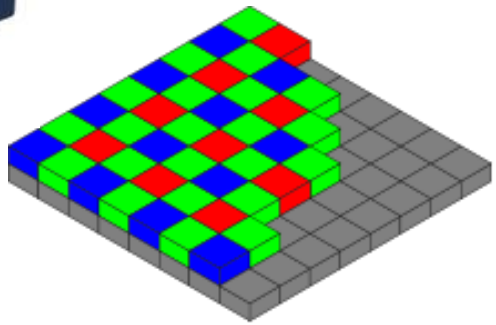


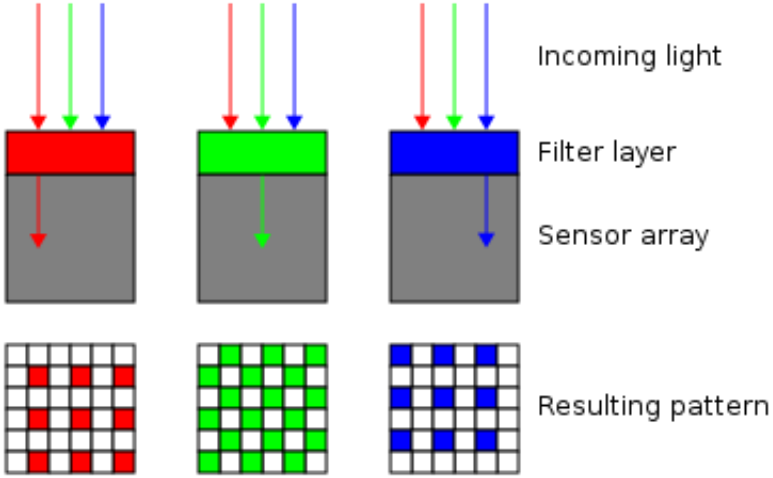
$$R = \int_{\lambda} E(\lambda)\, r_{Skin}(\lambda)\, f_R(\lambda)\, d\lambda$$

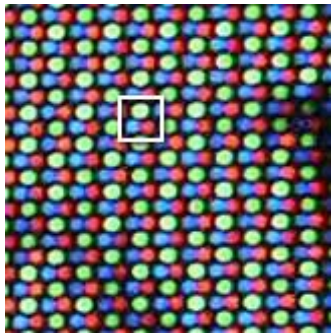
$$G = \int_{\lambda} E(\lambda)\, r_{Skin}(\lambda)\, f_G(\lambda)\, d\lambda$$

$$B = \int_{\lambda} E(\lambda)\, r_{Skin}(\lambda)\, f_B(\lambda)\, d\lambda$$

- Digital cameras often operate with RGB in a *Bayer filter* arrangement: *green* is given twice as many detectors as *red* and *blue* (ratio 1:2:1) in order to achieve higher luminance than chrominance resolution.
- The CCD sensor has a grid of red, green, and blue detectors arranged so that the first row is GBGBGBGB, the next is RGRGRGRG and that sequence is repeated in subsequent rows.





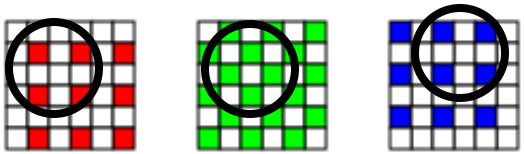


Three separate detectors each one receiving filtered red, green, or blue color ranges is an alternative to Bayer filter

Demosaicing algorithms

red green blue

Bayer filter samples



original

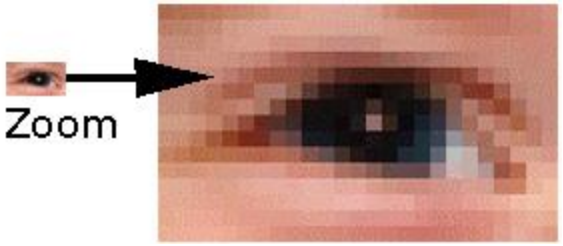
reconstructed after demosaicing

- Since, originally, a pixel only records one color out of three, it cannot determine the color of the reflected light. An algorithm is needed to estimate for each pixel the color levels for all color components, rather than a single component.

- To obtain a full color image demosaicing algorithms are used that interpolate a set of complete green, red, blue values at each point. This is done in-camera. For example, demosaicing can be done with bilinear interpolation the red value of a non-red pixel is computed as the average of the two or four adjacent red pixels, and similarly for blue and green….

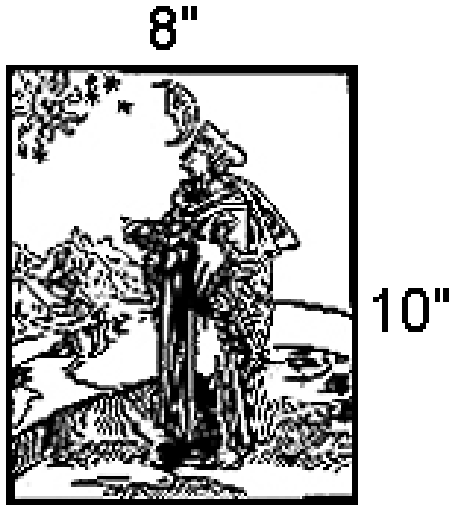
Image resolution, pixel dimensions

- *Resolution* is the density of the pixels in a given area. It is related to the ability to distinguish fine spatial detail. Pixels-per-inch (PPI) are common terms used to express resolution for digital images



- *Pixel dimensions* are the horizontal and vertical measurements of a given area in pixels. The pixel dimensions may be determined by multiplying both the width and the height of the area (in inches) by the ppi

2,400 pixels (8" x 300 ppi)
by
3,000 pixels (10" x 300 ppi)



Digital camera resolution, pixel dimensions

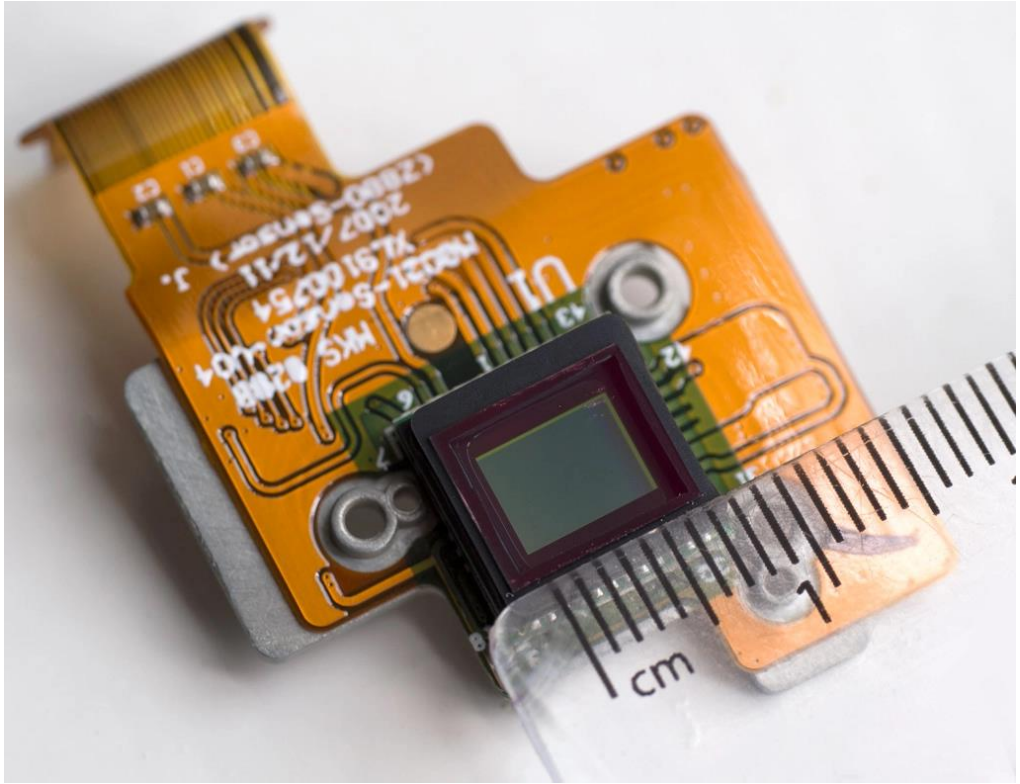
- In a digital camera, *resolution* (*pixel density*) is the number of pixels divided by the area of the camera sensor. A digital camera has *pixel dimensions*, expressed as the number of pixels horizontally and vertically that define its image resolution

- Digital cameras have a range of preset image resolutions which vary from camera to camera

Camera pixel dimensions	*Camera resolution* (*pixel density*)
640 x 480	low end
1216 x 912	1 Megapixel
1600 x1200	2 Megapixels, "high resolution."
2240 x1680	4 Megapixel cameras, the current standard
4064 x2704	11 Megapixels top-of-the-line digital camera
	Over 20 Megapixels high-end consumer cameras

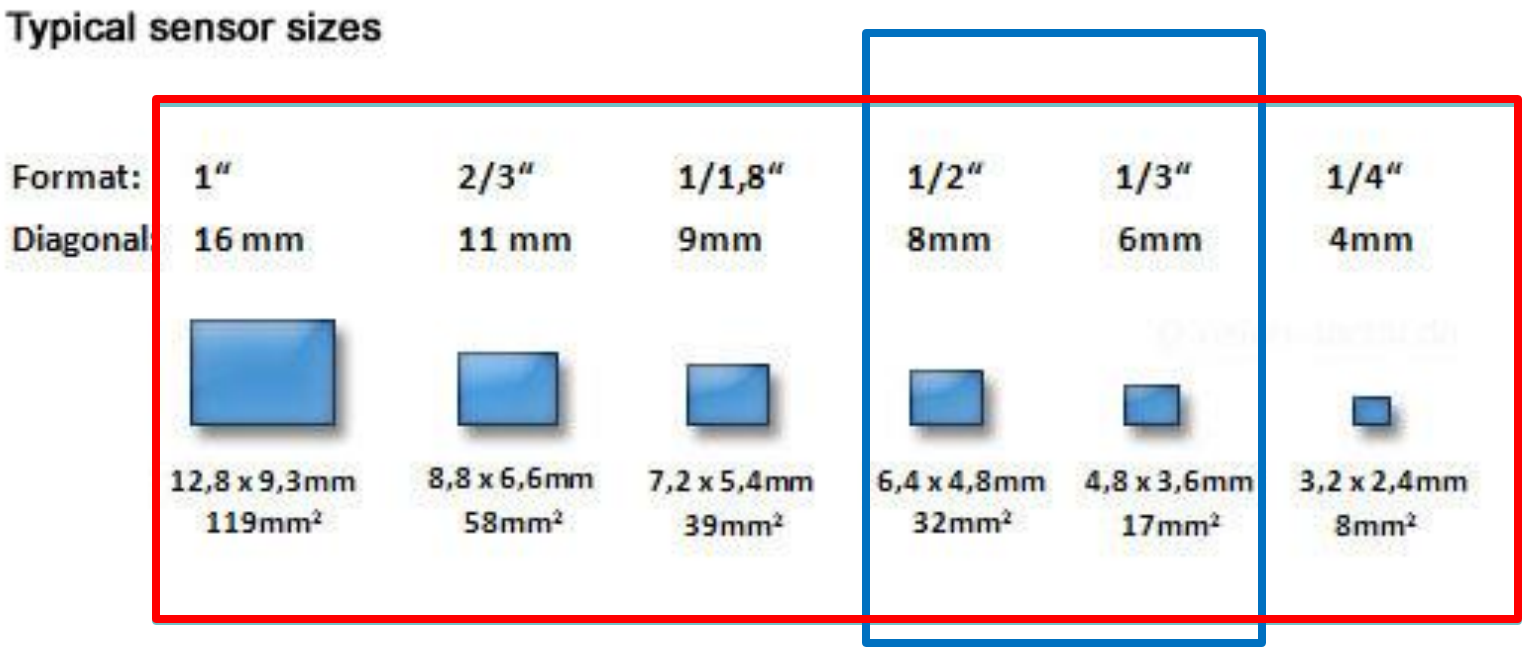
Camera sensor and pixel size

- Cameras have varyingly large sensors, depending on the camera and resolution used.
  As a general trend, in consumer cameras sensor and pixel sizes shrink in order to cut more and more sensors out of one wafer.



Camera sensor size

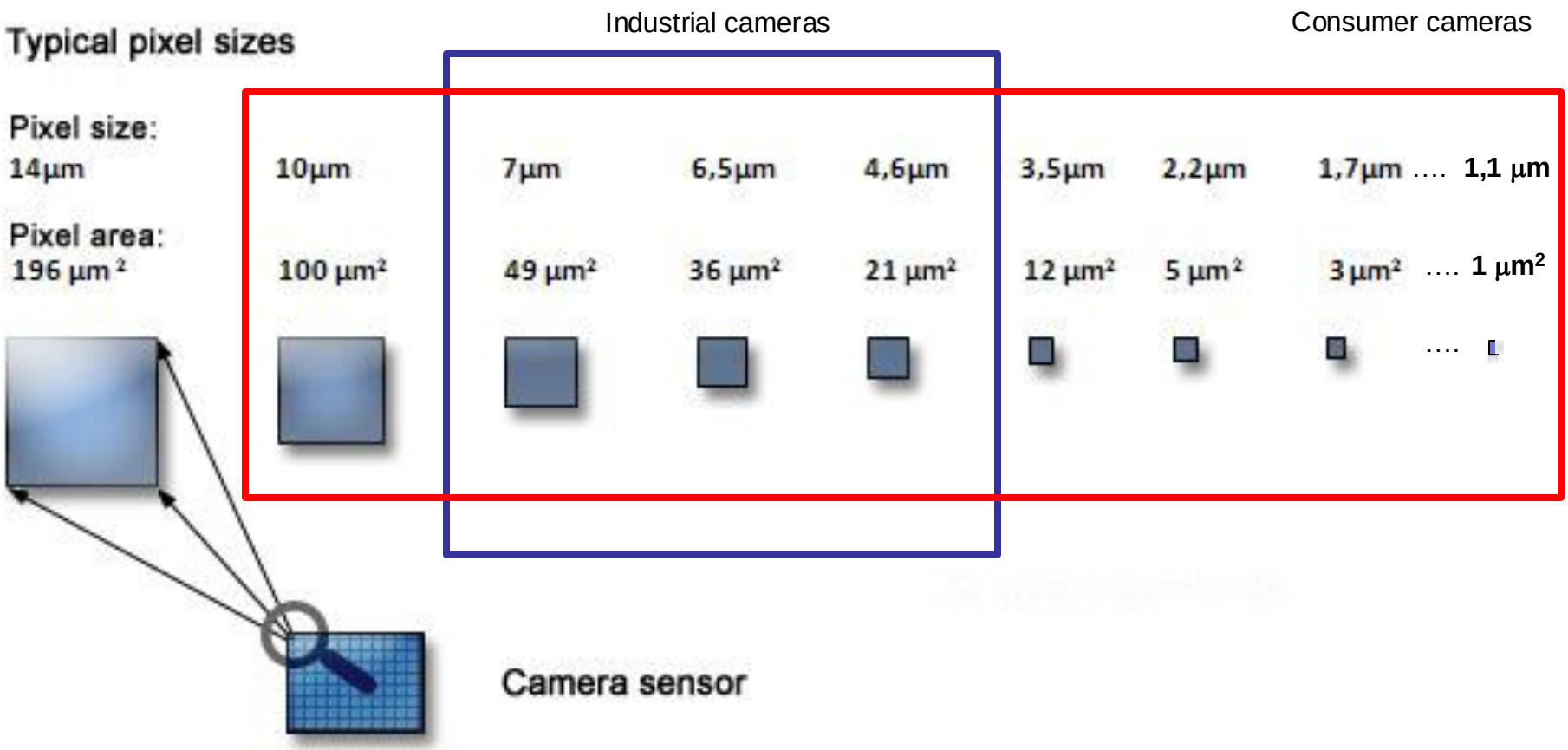
- In standard *consumer cameras* sensors range from 4 to 16 mm diagonal
- *Industrial cameras* have a 10 - 15 times larger light-active surfaces than consumer cameras. They usually use 6mm diagonal sensors for resolutions of 640 x 480 pixels and 8mm diagonal sensors with 1280 x 1024 pixels



Camera pixel size

- Consumer cameras have sensors with pixel sizes that range from 1.1 microns in the smallest smartphone sensor, to 8.4 microns in a Full-Frame sensor.

- Industrial cameras with resolutions up to 2 Megapixels normally have pixels of 4.6 to 7 µm



Camera resolution and pixel size

- Megapixel count is not an accurate reflection of a camera's quality. Factors to consider are also:
  - Amount of light
  - Amount of noise tolerated
  - Precision measurements to be conducted

- Smaller pixels in the same sized sensor record finer details but:
  - The smaller the pixels, the more light is required to capture an image. Smaller pixels are better when you have plenty of light. In case of industrial applications with short inspection times insufficient light can easily be a problem.
  - In case of little light, small pixels deliver noisier images than large pixels. Noise is disturbing for the application. For quality control this is absolutely inacceptable.
  - Many megapixels do not necessarily help. Small pixel structures require high-resolution lenses. Otherwise, blurred images with many pixels are created, but no real structure details

Infrared and thermal cameras

- Active IR systems use short wavelength infrared light to illuminate an area of interest.
- Some of the infrared energy is reflected back to a camera and interpreted to generate an image The radiation energy of each object varies depending on the variations of the values of the temperature of each object

- Thermal imaging systems use mid- or long wavelength IR energy. Thermal images are passive, and only sense differences in heat.
- The thermogram (the picture that is received) is a representation of the values of the mid-infrared radiation that the body of the object emits.

Thermal image heat measure

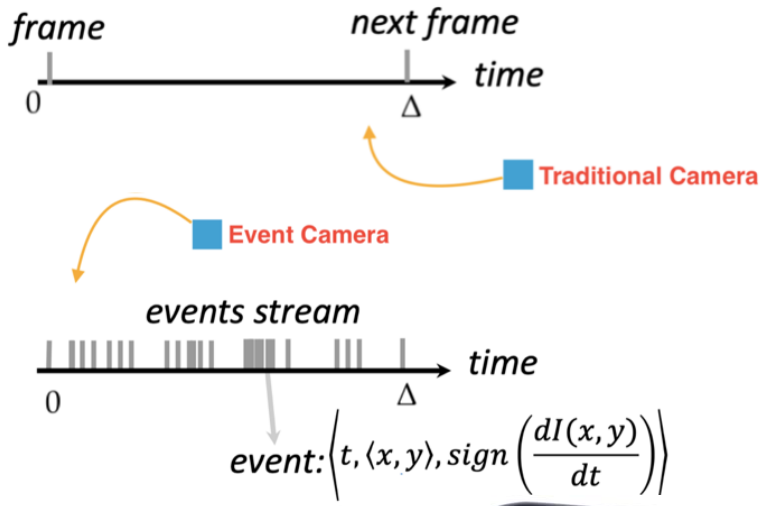
Inhaling

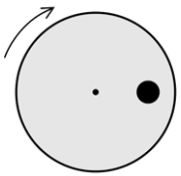
Exhaling

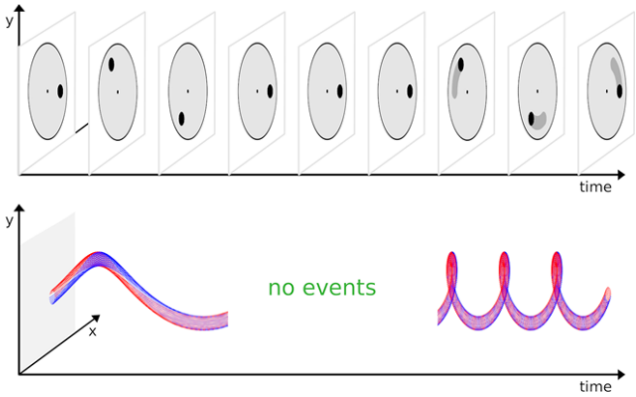
Event neuromorphic cameras

- An event camera is a bio-inspired sensor that works in a radically different manner compared to standard cameras: it measures only motion in the scene. On the market since 2016

- Instead of capturing a synchronous stream of frames, an event camera measures brightness changes (known as *events*) for each pixel.

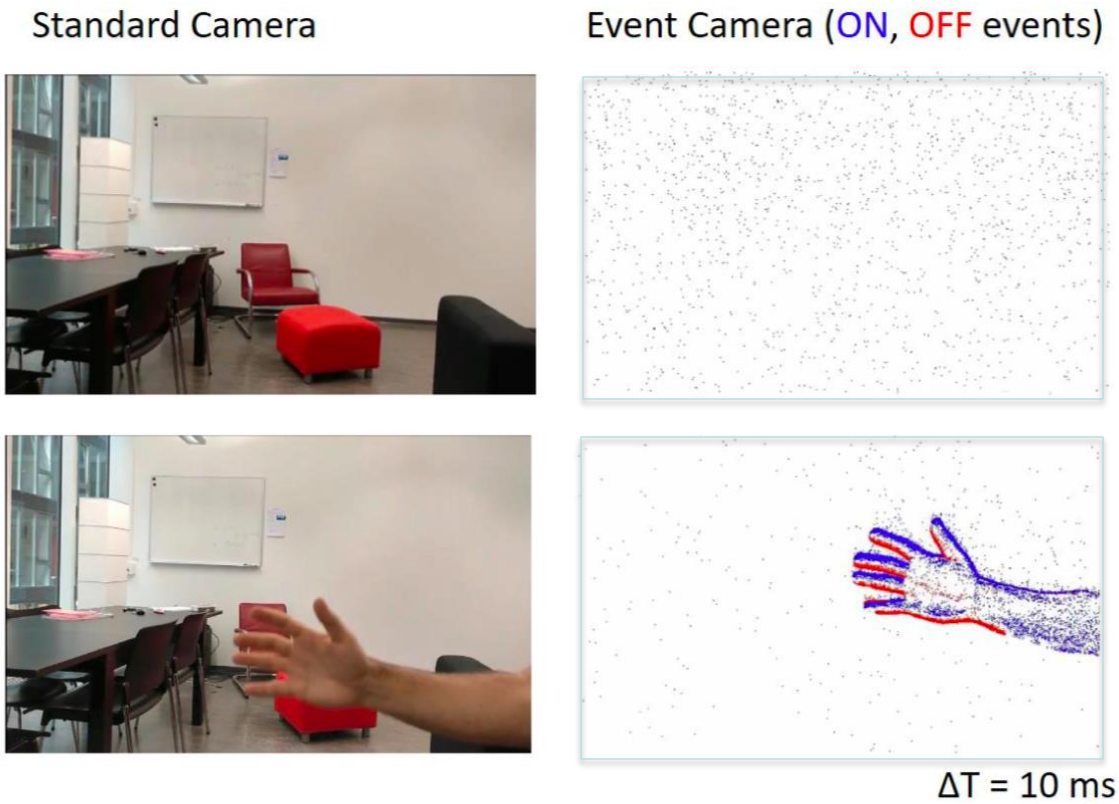






Event neuromorphic cameras

- Event cameras don't output frames but asynchronous events that are single pixel intensity changes at microsecond resolution (up to 100,000 fps).
  Similar behavior as hawks in nature….



Event cameras features

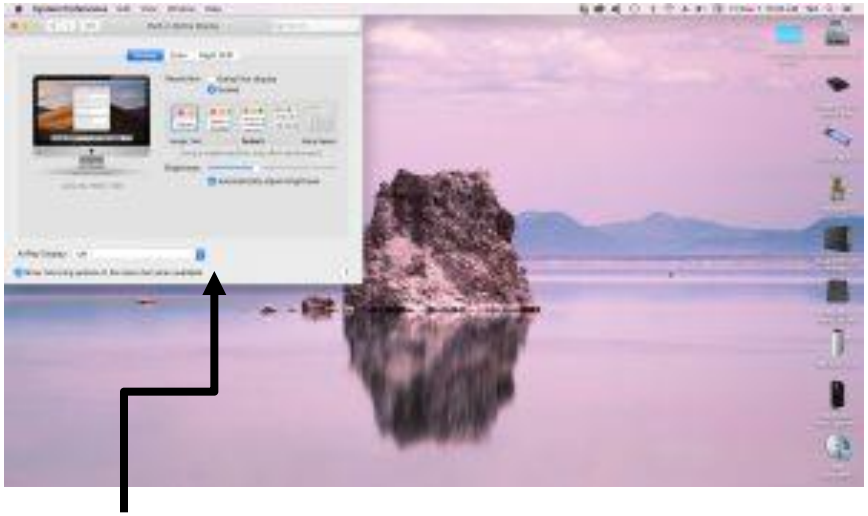
- Event cameras have a large potential for robotics and computer vision in challenging scenarios, such as high speed and high dynamic range.
  - High temporal resolution (in the order of microseconds)
  - Very high dynamic range (140 dB vs. 60 dB)
  - Low power consumption (mean: 1 mW instead of 1 W)
  - Low latency (in the order of microseconds)
  - No motion blur

Facial motion detection with event camera

Display resolution

- The resolution of a digital screen is the number of pixels in each dimension that are displayed.
- The size at which an image appears on the display depends on two things:
  - the pixel dimensions of the image
  - the display resolution of the screen
- *Native display resolution* is the published resolution of the screen. Most PCs also allow to change to a lower screen resolution

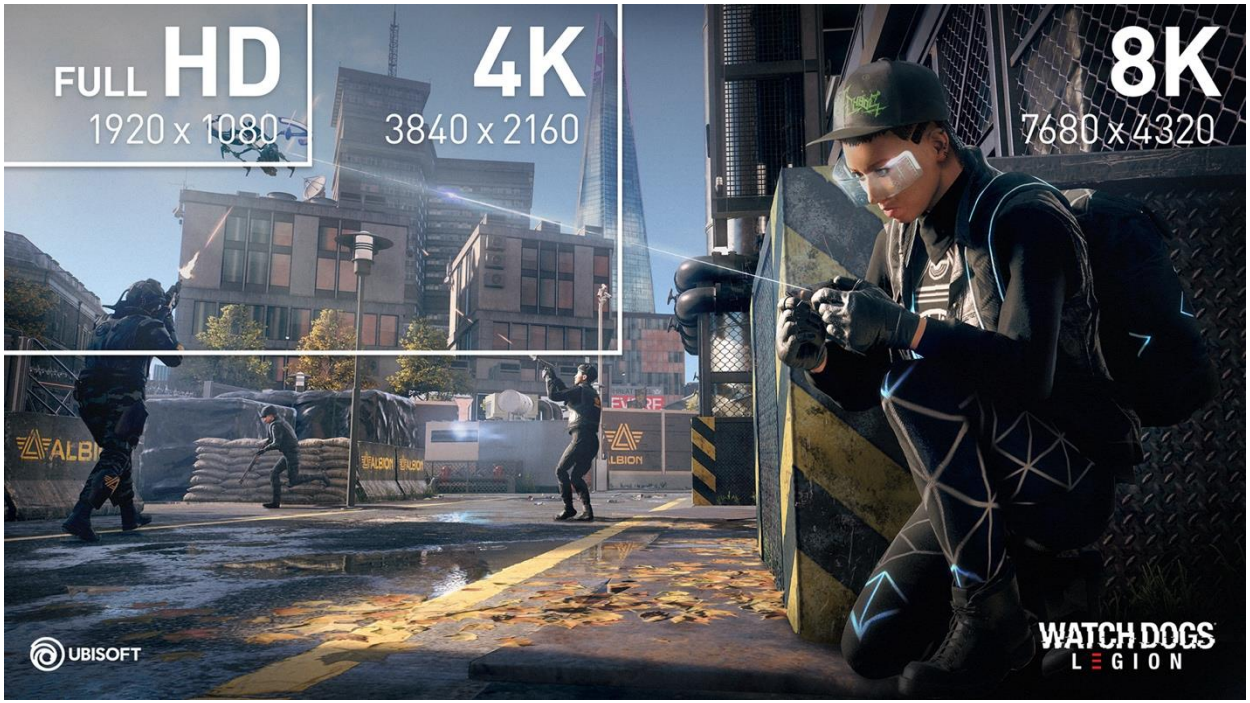
  With *native display resolution* each pixel in the image will take up exactly one pixel on the screen. For example, a 640x480 pixel image would fill a 640x480 pixel area of the screen



lowest screen resolution

largest screen resolution

Display resolution



- Displays are available at different resolutions:

HD	1280 x 720 pixels, 720p
Full HD	1920 x 1080 pixels, 1080p
2K	2048 × 1080 pixels
QHD, WQHD (2.5K Quad High Definition)	2560 x 1440 pixels, 1440p
UHD (Ultra High Definition)	3840 x 2160 pixels, 2160p
4K	4096 x 2160 pixels, 2160p
8K	7680 x 4320 pixels, 4320p

- Most high end monitors allow upscaling to convert low-resolution material (videos or images) into higher definition

Recommendations for web images

- When determining the correct size for an image intended for the web, only pixel dimensions count, i.e. how many pixels tall and how many pixels wide the image is.
  Resolution is meaningless on the web because the display device will determine its resolution.

- The image is resized in pixels at the size you want it to display. Devices may have different ratio than the image and it is will be cropped when viewed on web
- There are various cases to use different image sizes for different areas of the web site:

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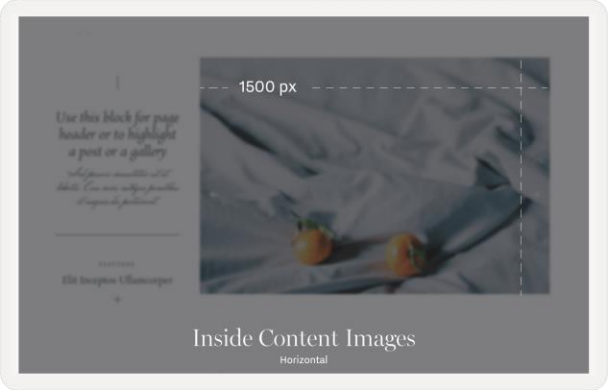
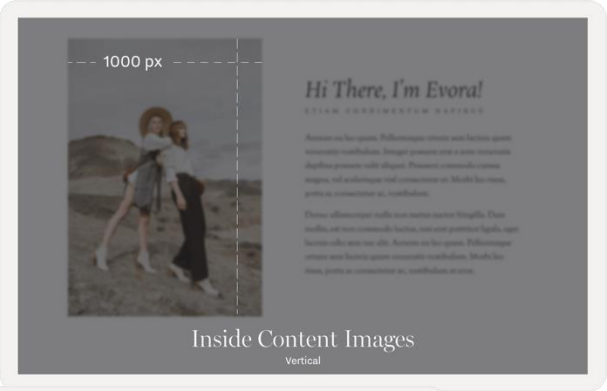
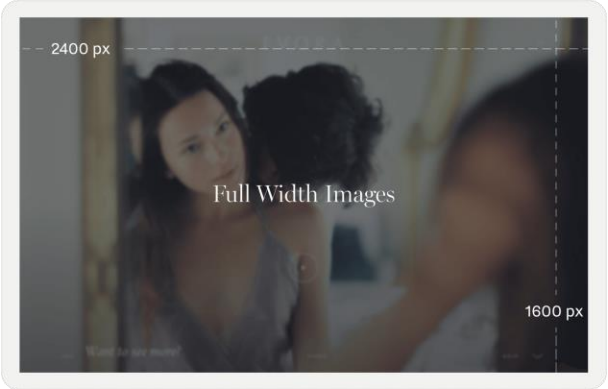
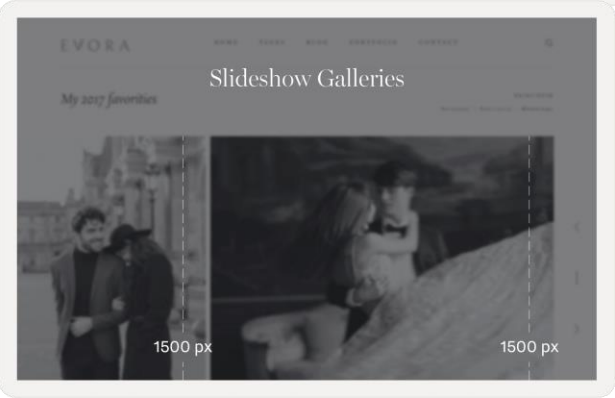
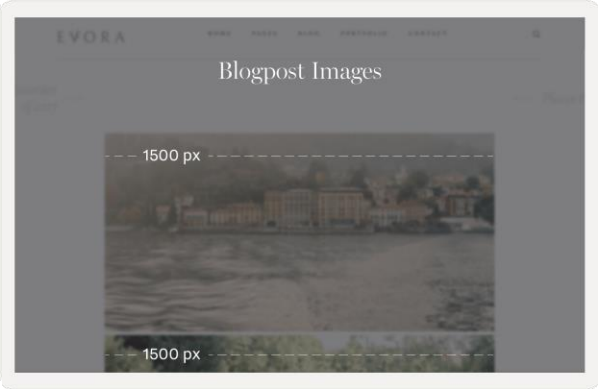




Courtesy flothemes.com







Color image representation

Color spaces

- A color space is a three-dimensional definition of a color system. The attributes of the color system are mapped onto the coordinate axes of the color space

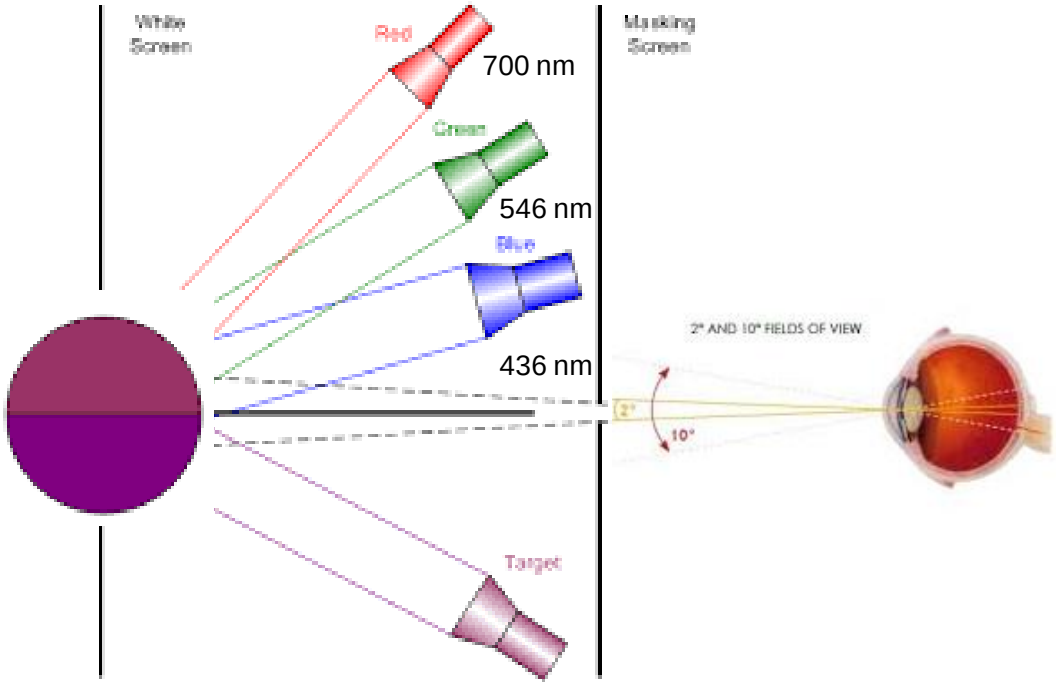
- The first color matching experiments were devised in late 1920s to characterize the relationship between the physical spectra and the perceived color

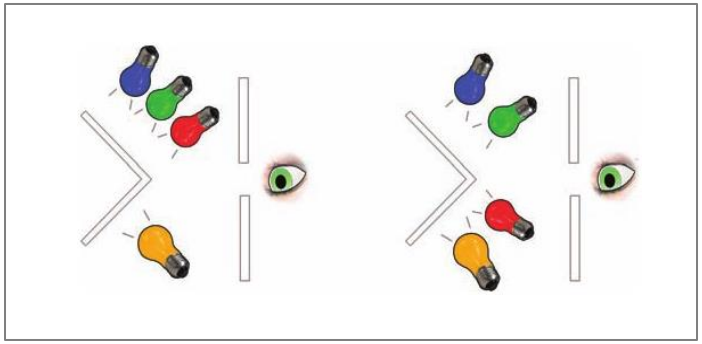
- Because of the need to have a color representation independent from the visualization system the CIE reference color space was defined. All color spaces are subsets of the CIE color space

CIE color matching experiment

- The CIE experiments used a circular split screen 2° in field of view size. On one side a test color was projected and on the other side, an observer-adjustable color was projected, that was a mixture of three monochromatic primaries at wavelengths of 700 nm (*red*), 546.1 nm (*green*) and 435.8 nm (*blue*), each with fixed chromaticity,but with adjustable brightness

- Not all test colors could be matched using this technique. When this was the case, a variable amount of one of the primaries was allowed to add to the test color. The amount of the primary added to the test color was considered to be a negative value

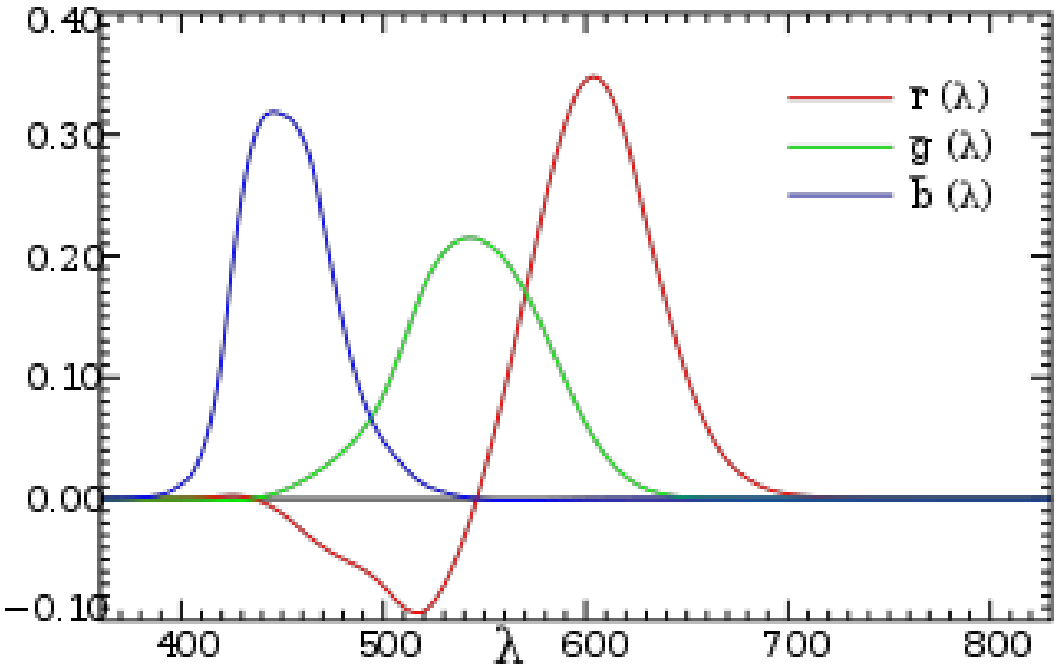




CIE RGB color space

- In 1931 CIE standardized the RGB color matching functions obtained using three monochromatic primaries roughly corresponding to color sensations of *red*, *green* and *blue*.
- These functions show the amount of primaries needed to match the test color at a given wavelength.

- The CIE RGB color space is one of many RGB color spaces, distinguished by a particular set of monochromatic primary colors.



$$R = \int_0^\infty I(\lambda)\,\bar{r}(\lambda)\,d\lambda$$

$$G = \int_0^\infty I(\lambda)\,\bar{g}(\lambda)\,d\lambda$$

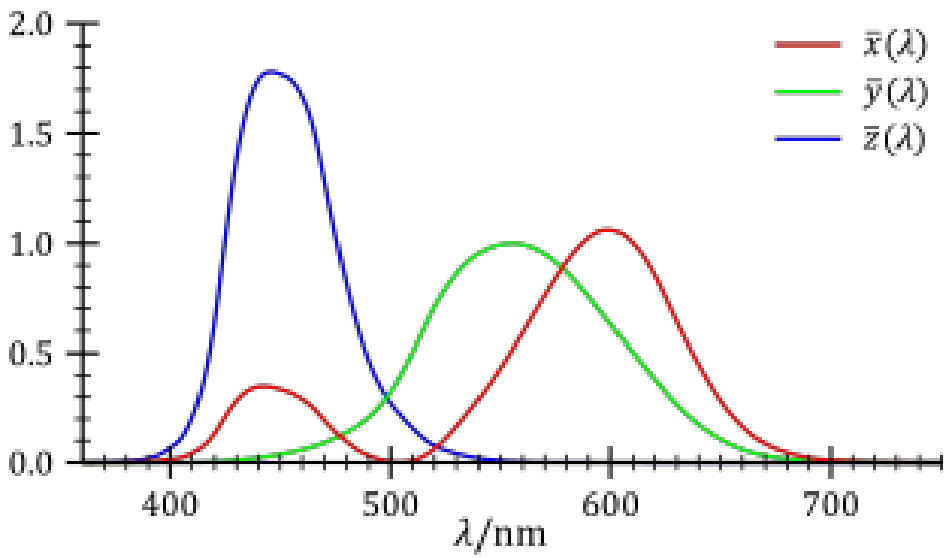
$$B = \int_0^\infty I(\lambda)\,\bar{b}(\lambda)\,d\lambda$$

CIE XYZ color space

- To avoid the presence of negative values of the CIE RGB matching functions, the Commission developed the CIE XYZ color space such that the new color matching functions were everywhere greater than or equal to zero.

- The CIE XYZ color space relates to the CIE RGB color space by a linear transformation. The transformation was designed such that:
  - Y was a measure of the *brightness* or *luminance* of a color,
  - Z was quasi-equal to *blue* stimulation,
  - X is a linear combination of the three cone response curves chosen to be nonnegative (roughly *red*).

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.4887180 & 0.3106803 & 0.2006017 \\ 0.1762044 & 0.8129847 & 0.0108109 \\ 0.0000000 & 0.0102048 & 0.9897952 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



$$X = \int E(\lambda)\,\bar{x}(\lambda)\,d\lambda$$

$$Y = \int E(\lambda)\,\bar{y}(\lambda)\,d\lambda$$

$$Z = \int E(\lambda)\,\bar{z}(\lambda)\,d\lambda$$

CIE xyY color space

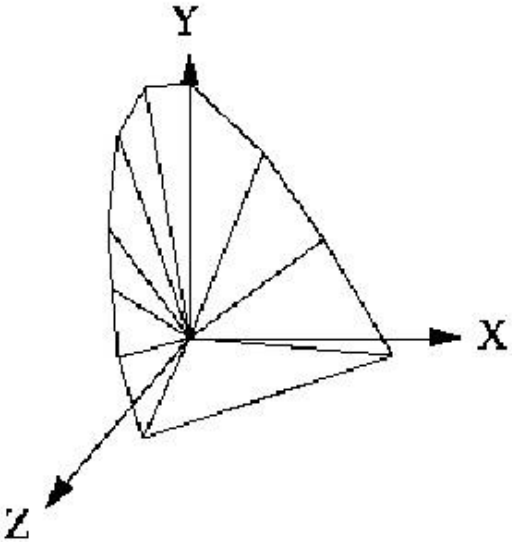
- Setting *Y* as luminance has the useful result that for any given *Y* value, the XZ plane will contain all possible chromaticities at that luminance.

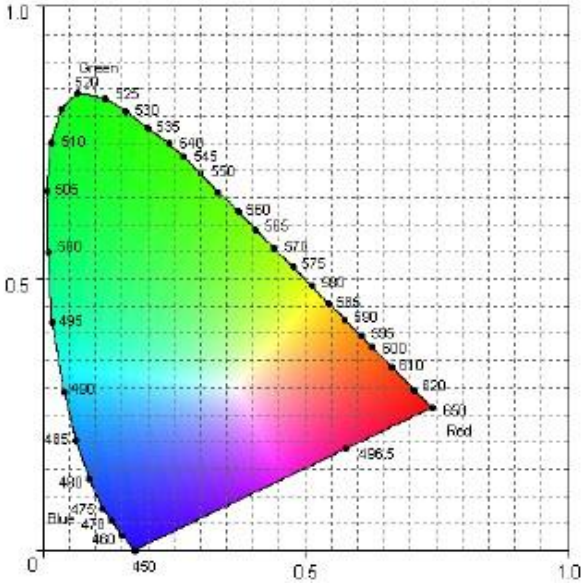
- By normalizing XYZ (i.e. dividing by X+Y+Z) derived values are obtained referred to as x,y,z .

$$x = \frac{X}{X+Y+Z} \quad y = \frac{Y}{X+Y+Z} \quad z = \frac{Z}{X+Y+Z} = 1 - x - y$$

- Being x+y+z = 1 the chromaticity of a color can be specified by two parameters x y, of the three normalized values and mapped into the CIE Chromaticity diagram

- The CIE Chromaticity diagram represents all the colors that are visible by the human eye with constant intensity equal to 1 (the brightest white supported by a color display). The degree of luminance is espressed by the Y coordinate of XYZ.





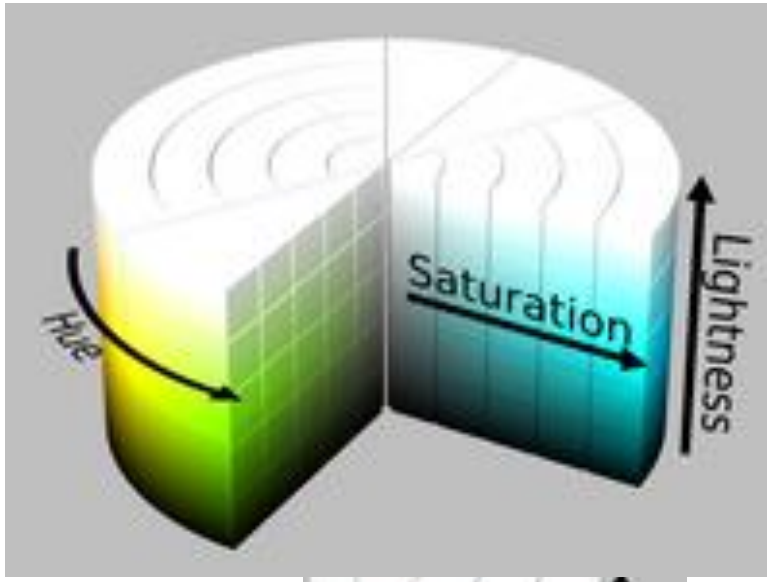
Different color spaces

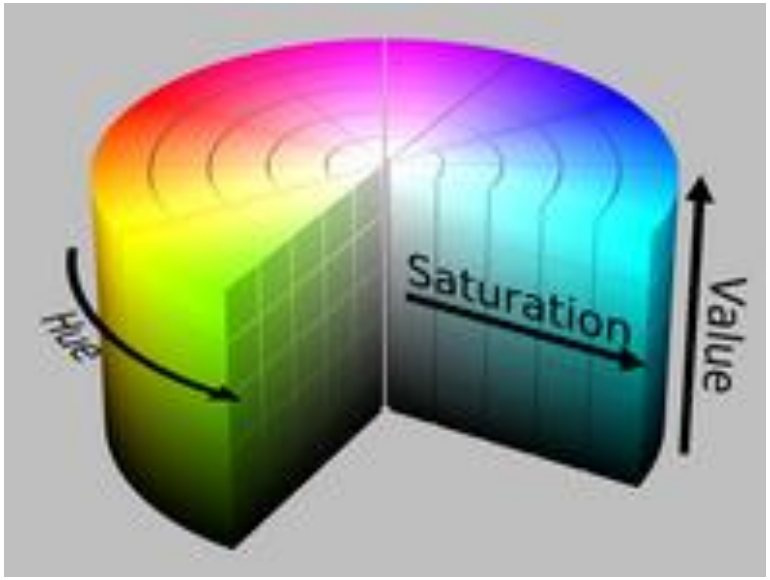
- Different color spaces exist. All mathematically derivable from the CIE reference color space: each has advantages and disadvantages for color selection and specification for different applications

Color spaces		**Applications**
CIE Colorimetric	XYZ	Colorimetric calculations
Device-oriented	Non-uniform spaces RGB, YIQ, YCC	Storage, processing, analysis, coding, color TV
	Uniform spaces L* a* b*, L* u* v*	Color difference evaluation, analysis, color management systems
Device-oriented and User-oriented	HSI, HSV, HSL, $I_1I_2I_3$	Human color perception, computer graphics
Computer vision	I1I2I3, c1c2c3	Invariance to highlights and shadows, respectively
Munsell		Human visual system

Different color spaces: HSL and HSV

- HSL and HSV are both cylindrical geometries, with hue, their angular dimension, with red primary at 0°, green primary at 120° and the blue primary at 240°, and then wrapping back to red. The central vertical axis comprises the neutral, achromatic, or gray colors ranging, white at top and black at bottom.

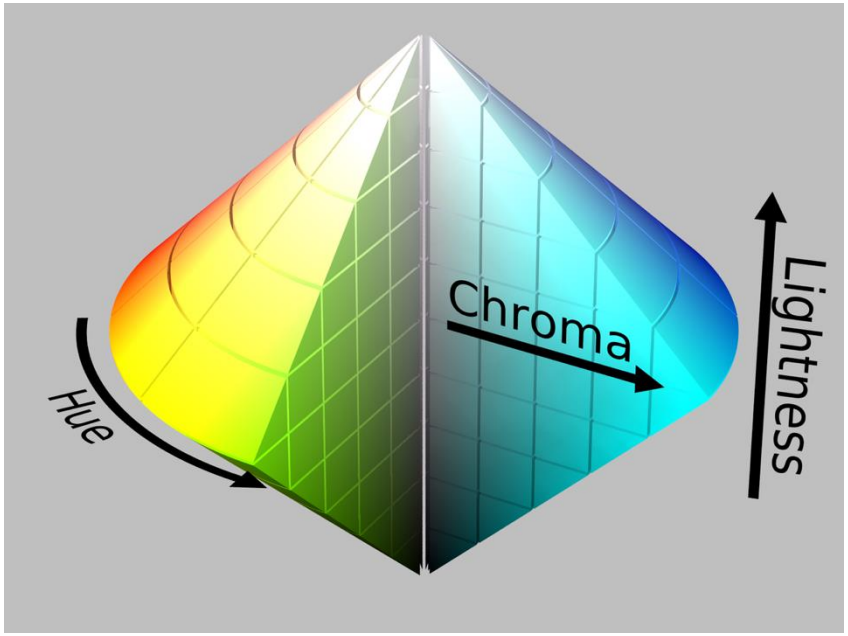


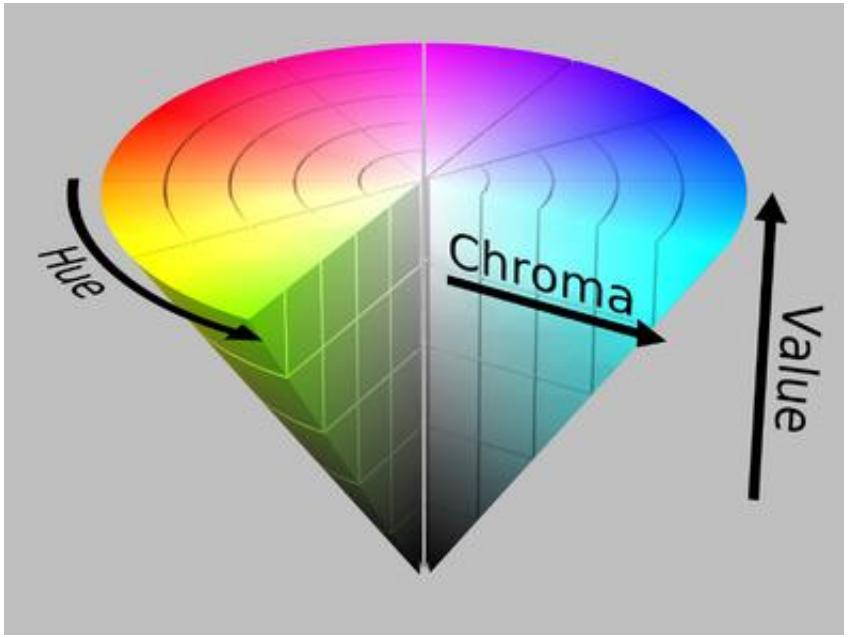


- Counterintuitively these two colors have the same saturation !

Different color spaces: HSL and HSV

- Using a conical representation is more intuitive, since it helps to differentiate very light or very dark near-neutral colors that would otherwise appear to have the same saturation (see external ring of the two cylinders).
- In this representation the radius is not anymore exactly a Saturation but rather a Chroma value, although the two terms are attributes related to the chromatic intensity. Many representations use the Saturation term anyway.



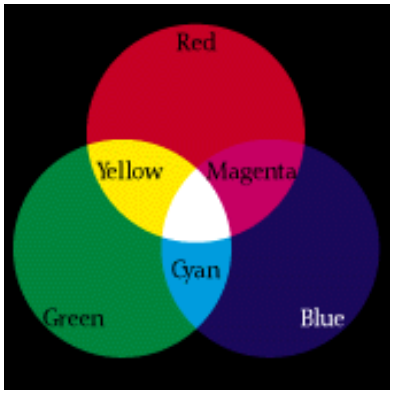


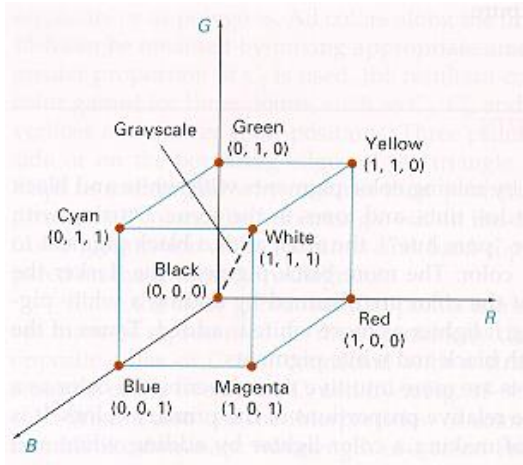
Different color spaces: HSL and HSV

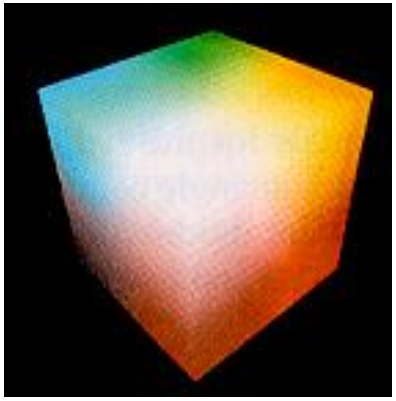
- HSL / HSV are often used in color pickers, e.g. in macOS

CIE RGB color space

- The CIE RGB color space is an additive color space, i.e. each color is obtained from the sum of the three primaries, and can be represented as a cube. The CIE RGB color space is related to XYZ by a linear transformation



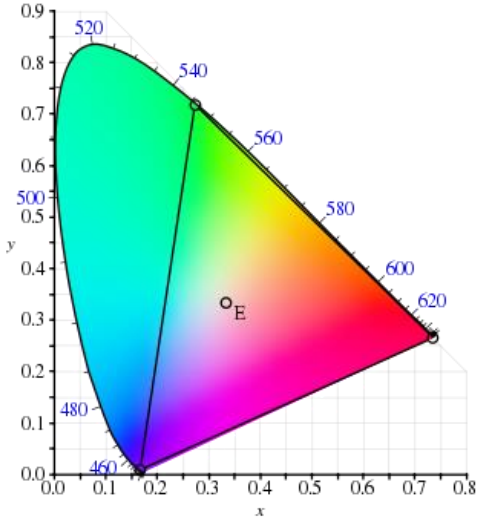




$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{vmatrix} 0.4887180 & 0.3106803 & 0.2006017 \\ 0.1762044 & 0.8129847 & 0.0108109 \\ 0.0000000 & 0.0102048 & 0.9897952 \end{vmatrix} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{vmatrix} 2.3706743 & -0.9000405 & -0.4706338 \\ -0.5138850 & 1.4253036 & 0.0885814 \\ 0.0052982 & -0.0146949 & 1.0093968 \end{vmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

CIE RGB gamut



RGB color spaces

- CIE RGB color space is one of the many RGB color spaces, each of which is distinguished by a particular set of monochromatic (single-wavelength) primary colors generated by the hardware. RGB is therefore hardware dependent

- The sRGB color space (created cooperatively by HP and Microsoft in 1996) is the most widely used in practice. It uses the same primaries as the ITU-R BT.709 primaries, standardized for studio monitors and HDTV and is the reference standard used for monitors, printers and on the Internet. LCDs, digital cameras, printers, and scanners all follow the sRGB standard

Color Space	**White Point**	**Primaries**					
		xR	yR	xG	yG	xB	yB
sRGB , HDTV (ITU-R BT.709),	**D65**	**0.64**	**0.33**	**0.30**	**0.60**	**0.15**	**0.06**
scRGB	D65	0.64	0.33	0.30	0.60	0.15	0.06
ROMM RGB	D50	0.7347	0.2653	0.1596	0.8404	0.0366	0.0001
Adobe RGB 98	D65	0.64	0.33	0.21	0.71	0.15	0.06
Apple RGB	D65	0.625	0.34	0.28	0.595	0.155	0.07
CIE RGB (1931)	E	0.7347	0.2653	0.2738	0.7174	0.1666	0.0089
……							

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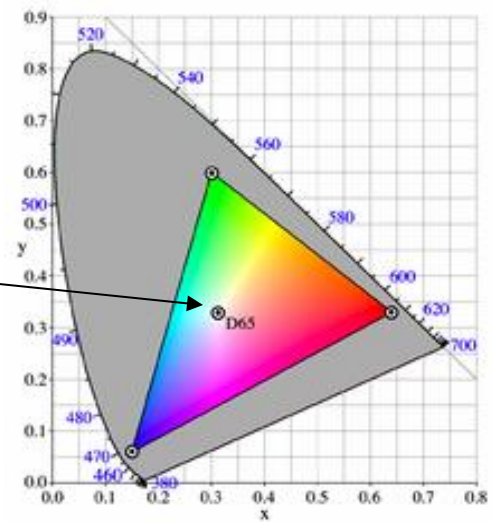
Standard daylight illuminant
White surface illuminated by average midday sun in western Europe/northern Europe



sRGB gamut

Note how sRGB has a smaller gamut than CIE RGB !

XYZ-RGB transformations

RGB Working Space	Reference White	RGB to XYZ [M]	XYZ to RGB $[M]^{-1}$
Adobe RGB (1998)	D65	0.5767309 0.1855540 0.1881852 0.2973769 0.6273491 0.0752741 0.0270343 0.0706872 0.9911085	2.0413690 -0.5649464 -0.3446944 -0.9692660 1.8760108 0.0415560 0.0134474 -0.1183897 1.0154096
AppleRGB	D65	0.4497288 0.3162486 0.1844926 0.2446525 0.6720283 0.0833192 0.0251848 0.1411824 0.9224628	2.9515373 -1.2894116 -0.4738445 -1.0851093 1.9908566 0.0372026 0.0854934 -0.2694964 1.0912975
Best RGB	D50	0.6326696 0.2045558 0.1269946 0.2284569 0.7373523 0.0341908 0.0000000 0.0095142 0.8156958	1.7552599 -0.4836786 -0.2530000 -0.5441336 1.5068789 0.0215528 0.0063467 -0.0175761 1.2256959
Beta RGB	D50	0.6712537 0.1745834 0.1183829 0.3032726 0.6637861 0.0329413 0.0000000 0.0407010 0.7845090	1.6832270 -0.4282363 -0.2360185 -0.7710229 1.7065571 0.0446900 0.0400013 -0.0885376 1.2723640
Bruce RGB	D65	0.4674162 0.2944512 0.1886026 0.2410115 0.6835475 0.0754410 0.0219101 0.0736128 0.9933071	2.7454669 -1.1358136 -0.4350269 -0.9692660 1.8760108 0.0415560 0.0112723 -0.1139754 1.0132541
CIE RGB	E	0.4887180 0.3106803 0.2006017 0.1762044 0.8129847 0.0108109 0.0000000 0.0102048 0.9897952	2.3706743 -0.9000405 -0.4706338 -0.5138850 1.4253036 0.0885814 0.0052982 -0.0146949 1.0093968
.....			
sRGB	D65	0.4124564 0.3575761 0.1804375 0.2126729 0.7151522 0.0721750 0.0193339 0.1191920 0.9503041	3.2404542 -1.5371385 -0.4985314 -0.9692660 1.8760108 0.0415560 0.0556434 -0.2040259 1.0572252
Wide Gamut RGB	D50	0.7161046 0.1009296 0.1471858 0.2581874 0.7249378 0.0168748 0.0000000 0.0517813 0.7734287	1.4628067 -0.1840623 -0.2743606 -0.5217933 1.4472381 0.0677227 0.0349342 -0.0968930 1.2884099

Truecolor representation

- The number of ones and zeros (bits) used to create each pixel denotes the depth of color you can put into your images.

- Truecolor is a method of representing and storing graphical image information. Truecolor defines 256 (2^8) shades of red, green, and blue for each pixel of the digital picture, which results in 256^3 or 16,777,216 (approximately 16.7 million) color variations for each pixel.

- Color depths of 30, 36, 48 bits per pixel are in use, also referred to as 10, 12, or 16 bits per RGB channel resulting in a billion or more colors

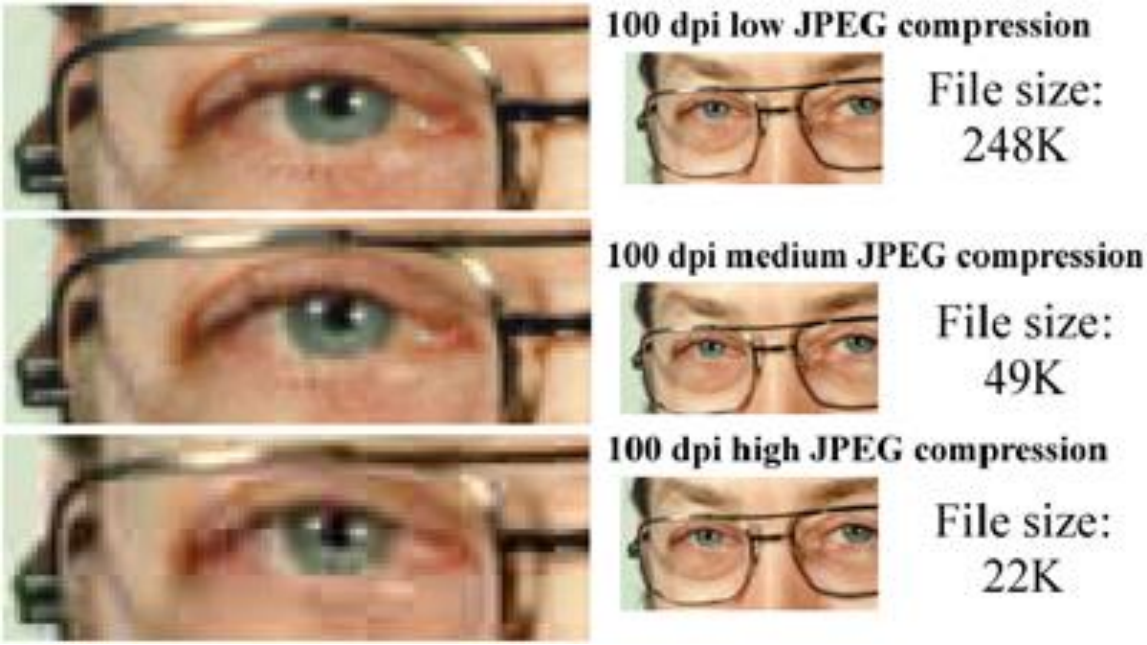
- In the absence of any other information, one can generally assume, that any 8-bit-per-channel image file or any 8-bit-per-channel image API or device interface can be treated as being in the sRGB color space.

Sample Length:	8								8								8							
Channel Membership:	**Red**								**Green**								**Blue**							
Bit Number:	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RGBAX	R. G. B. A. X																							
Sample Length Notation:	**8.8.8.0.0**																							

Image file size, compression

- File size is calculated by multiplying the area of the image (height x width) by the bit depth

  File Size = (pixel dimensions x bit depth) / 8

- Compression is used to reduce image file size for storage, processing, and transmission.

- All compression techniques abbreviate the string of binary code in an uncompressed image to a form of mathematical shorthand, based on complex algorithms.